

Effects of dietary factors and other metabolic modifiers on quality and nutritional value of meat.

A number of technologies that increase feed efficiency and lean tissue deposition while decreasing fat deposition have been developed in an effort to improve profitability of animal production. In general, the mode of action of these metabolic modifiers is to increase muscle deposition while often simultaneously reducing fat deposition. However, there have been some concerns that the focus on increasing production efficiency and lean meat yield has been to the detriment of meat quality. The aim of this review is to collate data on the effects of these metabolic modifiers on meat quality, and then discuss these overall effects. When data from the literature are collated and subject to meta-analyses it appears that conservative use of each of these technologies will result in a 5-10% (0.3-0.5kg) increase in shear force with a similar reduction in perception of tenderness. However, it should be borne in mind that the magnitude of these increases are similar to those observed with similar increases in carcass leanness obtained through other means (e.g. nutritional, genetic selection) and may be an inherent consequence of the production of leaner meat. To counter this, there are some other metabolic factors and dietary additives that offer some potential to improve meat quality (for example immuncastration) and it is possible that these can be used on their own or in conjunction with somatotropin, approved β -agonists, anabolic implants and CLA to maintain or improve meat quality.